

Import Data Using Transform Maps: Loading External Data into ServiceNow

Project Description:

This project uses ServiceNow's import tools to load data from external Excel/CSV files into ServiceNow tables quickly and accurately, replacing slow and error-prone manual data entry. The solution includes a Data Source, which uploads the source file, an Import Set table, which stages the raw rows, and a Transform Map, which maps the source fields to the target table using a coalesce field and transform scripts.

Built on the ServiceNow platform and tested on a Personal Developer Instance (PDI), the solution prevents duplicate records, cleans and validates data before it is saved, and skips and logs invalid rows instead of stopping the import. Access is controlled through ServiceNow roles and ACLs, and the Transform History makes it easy for administrators to review every import with confidence.

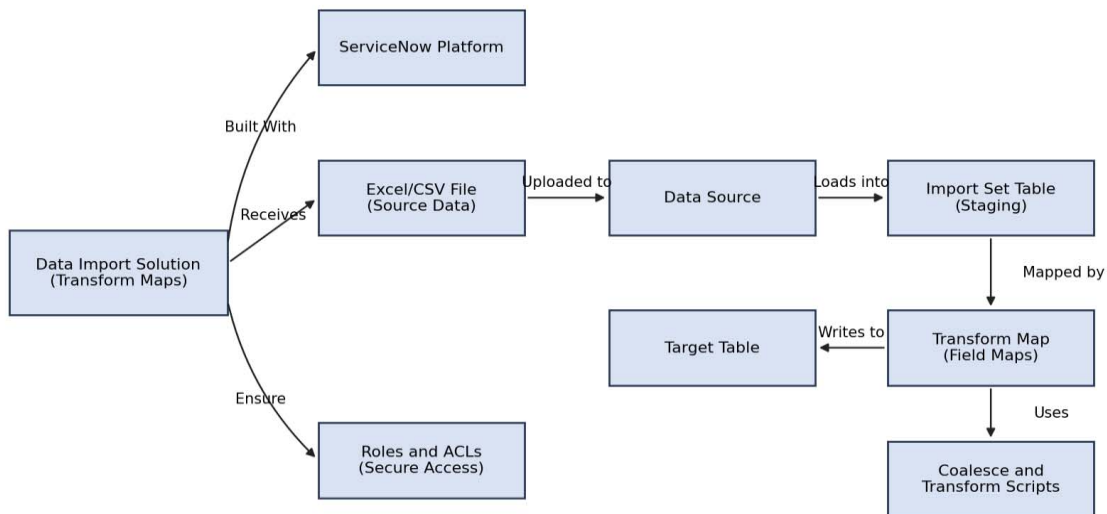
Scenarios

Scenario 1: An administrator receives an Excel file of new employee records. They create a Data Source, load the file into an Import Set table and run the Transform Map. The records appear in the target table within minutes, without any manual typing.

Scenario 2: The same file is imported again with updated email addresses. Because a coalesce field is set on a unique column, the existing records are updated instead of duplicated, so the table stays clean.

Scenario 3: A file contains rows with a missing mandatory value. The onBefore transform script skips those rows and logs an error. The administrator opens the Transform History, finds the skipped rows and fixes the source file.

Technical Architecture:



Description: The solution uses a simple three-layer architecture. The presentation layer is the ServiceNow platform UI, where the administrator works with the Data Source, Import Set and Transform Map forms. The logic layer is the Transform Map, which applies field mappings, the coalesce field and transform scripts. The data layer holds the Import Set staging table, the target table and the Transform History logs. Roles and ACLs control who can run imports.

(Note: The project stores its data in ServiceNow's own tables and uses no separate database, so an ER diagram is not needed.)

Pre-requisites:

- **ServiceNow Platform Knowledge:** Basic navigation of the ServiceNow instance, lists and forms.
- **Import Sets and Transform Maps:** Familiarity with Data Sources, Import Set tables and Transform Maps in the ServiceNow documentation.
- **JavaScript Skills:** Basic JavaScript for writing transform scripts.
- **Data Preparation:** Preparing a clean Excel or CSV file with a header row.
- **Update Sets:** Moving configuration between instances.
- **Development Environment Setup:** A free ServiceNow Personal Developer Instance from developer.servicenow.com.

Project Workflow:

Milestone 1: Environment Setup and Planning

- **Activity 1.1:** Get a ServiceNow Personal Developer Instance and log in as an admin.
- **Activity 1.2:** Prepare the Excel/CSV source file with a clean header row.
- **Activity 1.3:** Define the architecture of the import, from the file to the target table.
- **Activity 1.4:** Create an Update Set to capture all configuration changes.

Milestone 2: Data Source and Import Set

- **Activity 2.1:** Create a Data Source and attach the source file.
- **Activity 2.2:** Load the file rows into the Import Set staging table.

Milestone 3: Transform Map Development

- **Activity 3.1:** Create the Transform Map from the Import Set table to the target table.
- **Activity 3.2:** Map the source fields to the target fields.
- **Activity 3.3:** Set a coalesce field to prevent duplicate records.
- **Activity 3.4:** Write an onBefore transform script to clean and validate data.

Milestone 4: Running and Verifying the Import

- **Activity 4.1:** Run the transform and watch the import progress.
- **Activity 4.2:** Review the Transform History, Import Set rows and target table.
- **Activity 4.3:** Re-import the same file to confirm that the coalesce field works.

Milestone 5: Deployment

- **Activity 5.1:** Complete the Update Set and export it as XML.
- **Activity 5.2:** Import, preview and commit the Update Set on another instance.

Milestone 6: Conclusion

Milestone 1: Environment Setup and Planning

In this milestone, we prepare the ServiceNow instance and the source data, and plan how the data will flow from the file into the target table.

Activity 1.1: Get a ServiceNow Personal Developer Instance

- **Step 1:** Sign up for a free account at developer.servicenow.com.
- **Step 2:** Request a Personal Developer Instance (PDI) from the developer site.
- **Step 3:** Log in to the new instance as an admin using the credentials provided. An inactive PDI may hibernate, so wake it from the developer site if it does not load.

[Insert screenshot of the ServiceNow instance home page here]

Activity 1.2: Prepare the Source File

Prepare an Excel or CSV file with one header row and no blank rows. Keep column names short and consistent, because they become the field names in the Import Set table. An example layout is shown below.

user_id	first_name	last_name	email
U1001	Asha	Rao	asha.rao@example.com
U1002	Ben	Lee	ben.lee@example.com

Activity 1.3: Define the Architecture

- **Source:** An Excel/CSV file uploaded as an attachment to a Data Source.
- **Staging:** An Import Set table that holds the raw rows before any change is made to real data.
- **Transformation:** A Transform Map with field mappings, a coalesce field and transform scripts.
- **Target:** The ServiceNow table that receives the final records, such as the User table.

Activity 1.4: Create an Update Set

- **Step 1:** Go to System Update Sets > Local Update Sets and click New.
- **Step 2:** Give it a clear name, such as Import data using transform maps, and click Submit and Make Current.

Every Data Source, Transform Map and script created from now on is recorded in this Update Set, so it can be moved to another instance later.

Milestone 2: Data Source and Import Set

In this milestone, we create the Data Source that reads the file and load its rows into an Import Set staging table.

Activity 2.1: Create the Data Source

- **Step 1:** Go to System Import Sets > Administration > Data Sources and click New.
- **Step 2:** Enter a name and a new Import Set table label. Set Type to File, Format to Excel (.xlsx) or CSV, and File retrieval method to Attachment.
- **Step 3:** Set Header row to 1 and Sheet number to 1 for an Excel file, then click Submit.
- **Step 4:** Reopen the Data Source and use the paperclip icon to attach the source file.

[Insert screenshot of the Data Source form here]

Activity 2.2: Load Data into the Import Set Table

- **Step 1:** In the Data Source, click the Load All Records related link.
- **Step 2:** When the load finishes, open the Import Set Rows list and check that every row from the file appears.

At this point the data is only staged. Nothing has been written to the target table yet, so the load is safe to repeat.

[Insert screenshot of the Import Set Rows list here]

Milestone 3: Transform Map Development

Milestone 3 is the core of the project. In this milestone, we build the Transform Map that moves the staged rows into the target table, with field mappings, a coalesce field and a validation script.

Activity 3.1: Create the Transform Map

- **Step 1:** Go to System Import Sets > Administration > Transform Maps and click New.
- **Step 2:** Enter a name, set Source table to the Import Set table, and set Target table to the table that should receive the data.
- **Step 3:** Click Submit to save the map.

Activity 3.2: Map the Fields

- **Auto Map Matching Fields:** Use this related link to map source columns that already match target field names.
- **Manual mapping:** In the Field Maps related list, click New and choose a source field and a target field for each remaining column.

[Insert screenshot of the Field Maps list here]

Activity 3.3: Set the Coalesce Field

- **Step 1:** Open the field map for a column that is unique for every record, such as user_id or email.
- **Step 2:** Select the Coalesce check box and save.

When a row has the same value as an existing record, ServiceNow updates that record. Otherwise it inserts a new one. This is what prevents duplicates.

Activity 3.4: Write the Transform Script

Open the Transform Scripts related list, click New and set When to onBefore. The script runs once for every row before it is saved, so it can clean values and skip bad rows.

```
(function runTransformScript(source, map, log, target /*undefined onStart*/ ) {  
  
    // Clean the email value  
    if (source.u_email) {  
        target.email = source.u_email.toString().trim().toLowerCase();  
    }  
  
    // Skip rows with a missing mandatory value  
    if (source.u_user_id.nil()) {  
        ignore = true;  
        log.error('Skipped row: missing user ID');  
    }  
  
})(source, map, log, target);
```

- **Clean data:** trim() and toLowerCase() standardise the email value before it reaches the target table.

- **Skip bad rows:** Setting ignore to true skips the current row, and log.error() records the reason in the Transform History.

Milestone 4: Running and Verifying the Import

In this milestone, we run the Transform Map and check that the data landed correctly in the target table.

Activity 4.1: Run the Transform

- **Step 1:** Open the Import Set that was created by the load and click Run Transform.
- **Step 2:** Choose the Transform Map and confirm. Wait for the progress page to show that the transform is complete.

Activity 4.2: Review the Results

- **Transform History:** Go to System Import Sets > Transform History to see the start time, end time and any errors for each run.
- **Import Set Rows:** Check the state of each row, such as inserted, updated, skipped or error.
- **Target table:** Open the target table list and confirm that the imported records are present with the cleaned values.

[Insert screenshot of the Transform History here]

Activity 4.3: Test the Coalesce Field

Load and transform the same file a second time. The Import Set rows should show updated instead of inserted, and the target table should not contain any duplicate records.

Milestone 5: Deployment

In Milestone 5, the focus is on moving the finished import configuration to another ServiceNow instance using the Update Set created in Milestone 1.

Activity 5.1: Export the Update Set

- **Step 1:** Go to System Update Sets > Local Update Sets and open the Update Set.
- **Step 2:** Set its state to Complete, then click Export Update Set to download the XML file.

Activity 5.2: Import the Update Set on the Target Instance

- **Step 1:** On the target instance, go to System Update Sets > Retrieved Update Sets and choose Import Update Set from XML.
- **Step 2:** Upload the XML file, open the loaded Update Set, and click Preview Update Set.
- **Step 3:** Resolve any preview problems, then click Commit Update Set.

Important Notes:

- Attach the source file to the Data Source again on the new instance, because attachments are not carried in the Update Set.
- Make sure the target table and its fields exist on the new instance before committing.
- Test the import with a small file first to confirm that everything works.

Exploring the ServiceNow Pages:

Data Source Page:

[Insert screenshot of the Data Source page here]

Description: The Data Source form defines where the data comes from. It holds the file type, the file retrieval method, the header row and the Import Set table name. The attached Excel/CSV file and the Load All Records link are on this page.

Transform Map Page:

[Insert screenshot of the Transform Map page here]

Description: The Transform Map form links the Import Set table to the target table. Its related lists show the field maps, the coalesce settings and the transform scripts.

Transform History Page:

[Insert screenshot of the Transform History page here]

Description: The Transform History lists every import run with its start and end times and the number of inserted, updated, skipped and error rows. Administrators use it to confirm a successful import and to find the cause of any failed rows.

Conclusion:

This project shows how Data Sources, Import Set tables and Transform Maps can load external data into ServiceNow quickly and accurately. A coalesce field prevents duplicate records, and transform scripts clean and validate each row before it is saved. Invalid rows are skipped and logged instead of stopping the import. Moving the setup between instances with an Update Set keeps it easy to repeat and maintain. Looking ahead, the solution can be extended with scheduled imports for repeat loads, email notifications when an import has errors, and support for other data sources such as JDBC or REST. With these improvements, the import can grow from a one-time load into a dependable, automated data pipeline for teams of all sizes.